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Master's Thesis

Data Consistency Evaluation for Reaction Model Optimization

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Thank somebody if appropriate.

Abstract

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2.1 Background

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2.2 Chemical Kinetics Database

TUMKIN, a chemical kinetic data infrastructure currently under development aims to incorporate databases and tools to facilitate analysis and optimization of reaction mechanisms of wide range of fuels. The idea is based on Process Informatics Model (PrIMe) created by Frenklach and coworkers which was a data infrastructure concentrating on data management, data uncertainty, and data consistency. [1] However PrIMe website is not active. TUMKIN aims to revive and expand on PrIMe. A similar effort is RESPECTH [2]. Their database contains direct and indirect experimental measurements for various fuels. Relevant for this thesis are the indirect shock tube ignition delay time measurements for Ethylene, Syngas and Hydrogen. [3][4][5] They also share computer programs for various tasks around analysis and optimization of kinetics. Reaction Mechanism Generator (RMG) is an automatic chemical reaction mechanism generator that constructs kinetic models composed of elementary chemical reaction steps using a general understanding of how molecules react. RMG database contain kinetics information for many reactions from various sources and datasets. [6]

[7]Pelucchi et al. describe a shift in combustion chemistry from handmade validation to the development of ”methods able to automatically validate the generated models”. Key aspects of their discussion include: +1 Need for Automation: The ”many-model” problem—where numerous mechanisms representing the same fuels use different rate constants and pathways—creates an ”increased need of comparisons with experimental data to improve the reliability and performances of the models”. Validation Approach: Validation is defined as the ”match between such amount of experiments and models,” where ”model estimations are compared with measurements obtained in defined conditions”. Quantitative Methodology: They propose a ”quantitative, comprehensive approach to estimate the agreement between

models and experimental data" that accounts for the "distance between measurements and predictions," "shapes of the interpolating curves," and "possible shifts on the independent variable". Curve Matching: The paper details a "Curve Matching approach for the automatic assessment of kinetic models" which is intended to include "experimental error and define normalization processes" to ensure applicability across different models and conditions. +1 Data Management: The authors emphasize that "effective algorithms and codes to compare models and measurements are required" alongside standardized data formats (such as RKD or YAML) to ensure "consistency, accountability, traceability, and accuracy".

2.3 Uncertainty Quantification in Combustion

Uncertainty quantification (UQ) is the science for quantification and minimization of uncertainties with computer experiments and simulation. [8] The approach to developing detailed kinetic models (mechanisms) of fuel combustion involves compiling a set of elementary reactions whose rate parameters may be determined from individual rate measurements, reaction-rate theory, or a combination of both. These methods have uncertainties. A widely used method is sensitivity analysis. UQ is closely related to sensitivity analysis, and in many ways, UQ is a natural extension of sensitivity analysis. A basic objective of UQ is to analyze the uncertainty in the model response because of model parameters being uncertain, including forward uncertainty quantification and propagation and an inverse problem leading to model uncertainty constraining. When treated as a Bayesian inference problem, UQ aids the development of predictive kinetic models by constraining models with data and yielding estimates for the confidence of a model to make predictions outside of the thermodynamic regimes where the model has been tested. [8]

2.4 Uncertainty of reaction mechanisms

Reaction mechanisms employed in chemical kinetics models include information about the relevant species, reactions, reaction rates and the corresponding uncertainties of these reaction rates. Reactions are modeled with the Arrhenius equation with k being the reaction rate coefficient (RRC). In the literature, Arrhenius coefficients (A,n,Ea) are fitted to experimental data with a regression model and evaluated with expert opinion. Or they are calculated from fundamental calculations. Therefore, different literature sources can report different reaction rate coefficient k(T) values for a given reaction. The uncertainty factor f is a logarithmic measure of the plausible spread of k(T) around its recommended value:

$$k(T) = AT^n \exp\left(-\frac{E_a}{T}\right) \qquad (\mathrm{cm^3\,mol^{-1}\,s^{-1}},\ \mathrm{K}) \tag{2.1}$$

To perform analysis or optimization of combustion mechanisms, it is important to quantify the uncertainty of its parameters. For every reaction there are many literature sources reporting varying parameter sets. The most comprehensive source of a database compiling such sources is NIST Kinetics Database. In this thesis, to quantify the uncertainty of Arrhenius expressions, literature sources are collected from RMG database. For every active reaction, nonlinear least squares fit is performed using the selected sources from the database. Fit is performed on the following regression model:

$$k(T) = 10^{\log_{10}(A)}\, T^n \exp\left(-\frac{E_a}{T}\right) \tag{2.2}$$

Parameter standard deviations $s\left(\log_{10}(A)\right), s\left(n\right), s\left(E_a\right)$ are used to calculate lower and upper uncertainty bounds for rate coefficients.

$$k_{\mathrm{low}}(T) = 10^{\log_{10}(A) - s(\log_{10} A)}\, T^{n - s(n)} \exp\left(-\frac{E_a + s(E_a)}{T}\right) \tag{2.3}$$

$$k_{\mathrm{up}}(T) = 10^{\log_{10}(A) + s(\log_{10} A)}\, T^{n + s(n)} \exp\left(-\frac{E_a - s(E_a)}{T}\right) \tag{2.4}$$

The uncertainty parameter f is defined as:

$$f(T) = \log_{10}\left(\frac{k_{\mathrm{up}}(T)}{k_0(T)}\right) = \log_{10}\left(\frac{k_0(T)}{k_{\mathrm{low}}(T)}\right) \tag{2.5}$$

Where k0 is the best fit of the rate coefficient, klow and kup are the lower and the upper bounds respectively [9]. Rate coefficients are then sampled from the interval $\left[k_0\left(T\right)/10^f\left(T\right), k_0\left(T\right) * 10^f\left(T\right)\right]$. Which preserves the shape of the k(T) curve and only scales it. It is also possible to use the joint multivariate distribution of Arrhenius parameters (A,n,Ea) sample reaction rate coefficients. Covariance matrix However, authors rarely report covariance matrix or parameter standard deviations. When the covariance matrix is not available, often only preexponential factor A is varied to achieve the prescribed uncertainty band on $k\left(T\right)$ [10].

2.5 Sensitivity Analysis

Local sensitivity analysis is a method where the response of the model to small parameter changes close to nominal values is explored. Normalized first-order sensitivity coefficients are commonly applied in combustion to provide kinetic insight into the model and its strengths and weaknesses with respect to experimental data.[11]

2.6 Surrogate Models

In the B2BDC framework, surrogate models are constructed using a solution-mapping approach over the prior uncertainty region. Active parameters are first identified through impact-factor screening and the surrogate is built only in this reduced parameter space. Training data are generated using space-filling designs, and the original model is evaluated at each design point. Polynomial, quadratic, or rational quadratic surrogate models are then fitted by minimizing either $L_2 or L_\infty$ norms. Rational quadratic surrogates are used when polynomial models are insufficiently accurate, and their coefficients are obtained through semidefinite programming [12].

2.7 Consistency Analysis

Given a reaction mechanism, chemical kinetics simulation software like Cantera**cantera** can simulate combustion, and predict properties like ignition delay time(IDT), laminar flame speed(LFS), etc. These properties are also measured in real life combustion experiments, and reported together with quantified uncertainties. A good reaction model should be able to predict these properties accurately under same operating conditions as the physical experiment(temperature, pressure, stoichiometric ratio, mixture content). Consistency analysis in this context are methods to validate simulation and experimental results against each other by finding a suitable set of model parameters that best explain the experiment data within their uncertainty intervals. If some data can't be explained by to model, this can suggest either the data is bad, or the model is bad.

2.7.1 Misc

ralph c. smith book check the residuals of model v data for given mechanism for structure. If residuals are biased, model doesnt capture everything. solution 1: add an model discrepency term. fit it as a polynomial. solution 2 : (bayes land)

if model descrepency is due to shortcomings of experiments(i.e scneario dependent) we would expect it to work best when modeled as different functions for different QoIs. however if it is a shortcoming of the model(ie. missing reactions) a single function(e.g polynomial) fit or a single prior distrubition of the discrepency should be enought to partially compensate.?? at least me thinks thta way.

Shortcomings of previous work can be due to model discrepancies. Reaction mechanism does not contain all chemical reactions relevant to the combustion processs of interest. Because omitting reactions in the mecahnisms makes the computanations easier and and if there a re reactions that are not tknow to be in the combustion process it wouldn't be in there any ways. So we assume it is not possible to improve the model to compenstate for the discrepancy. Instead I will develeop a statistical method to try to partially identify and partially compensate this discrepancy. One such method

[13]. Model discrepency. Kennedy and O'Hagan (2001) provide a foundational Bayesian framework for the calibration of computer models that explicitly accounts for various sources of uncertainty, most notably model inadequacy. In their formulation, they recognize that "no model is perfect" and define the discrepancy as follows: They further specify that "model inadequacy [is] the difference between the true mean value of the real world process and the code output at the true values of the inputs". This term captures the reality that "even if there is no parameter uncertainty... the predicted value will not equal the true value of the process". To address this, their Bayesian technique "attempt[s] to correct for any inadequacy of the model which is revealed by a discrepancy between the observed data and the model predictions from even the best-fitting parameter values". They implement this by "modelling both the computer code output and model inadequacy" using Gaussian processes.

[14] Is a review of deep learning approaches to uncertainty quantification from various fields and methods. This study provides a comprehensive review of recent advances in uncertainty quantification (UQ) methods used in deep learning and investigates the application of these methods in reinforcement learning. It highlights how UQ methods used along different machine learning and deep learning techniques can significantly increase the reliability of their results. By reviewing most recent articles published on quantifying uncertainty in

artificial intelligence, this paper serves as a comprehensive guide to navigating this fast-growing research field.

2.7.2 Anomaly Detection / Gross Error Detection

[15]"Data cleansing offers a better data quality which will be a great help for the organization to make sure their data is ready for the analyzing phase. This paper reviews the data cleansing process, the challenge of data cleansing for big data and the available data cleansing methods. Data cleansing process mainly consists of identifying the errors, detecting the errors and corrects them. Ensuring the accuracy and relevancy of data will drive the business a step ahead from their competitors."

[16]"The technique developed at NRC for gross-error detection is composed of four steps. Each step deals with errors of a certain magnitude and nature such as errors in point identification or errors in the control points. Non-statistical approaches are employed for the first three types of gross errors and a rigorous statistical method (data snooping) for the fourth type. The final step, which deals with small observation errors of magnitudes larger than the random range, consists in applying a sensitive statistical test (data snooping). Using the OEEPE test blocks, which include all types of errors (intentionally introduced), the above procedure was applied and proved to be effective."

[17]Real-world datasets often contain anomalous cases, also known as outliers. A useful tool for this purpose is robust statistics, which aims to detect the outliers by first fitting the majority of the data and then flagging data points that deviate from it. To avoid the masking and swamping effects of classical methods, the goal of robust statistics is to find a fit which is close to the fit we would have found without the outliers. Outliers are then identified by their large 'deviation' (e.g., its distance or residual) from that robust fit.+3In the multivariate case, a robust measure of location $\hat{\mu}$ and scatter $\hat{\Sigma}$ can be used to compute robust distances for any point x:

$$RD(x, \hat{\mu}, \hat{\Sigma}) = \sqrt{(x - \hat{\mu})^T \hat{\Sigma}^{-1} (x - \hat{\mu})} \tag{2.6}$$

Robust estimates can be obtained via the Minimum Covariance Determinant (MCD) method, which looks for those h observations in the dataset whose classical covariance matrix has the lowest possible determinant. Alternatively, in a linear regression setting, the Least Trimmed Squares (LTS) estimator identifies outliers by minimizing the sum of the h smallest squared residuals:

$$\text{minimize} \sum_{i=1}^{h} (r^2)(i) \tag{2.7}$$

where $(r^2)(1) < (r^2)(2) < ... < (r^2)(n)$ are the ordered squared residuals.

[18]As established in the survey by Chandola et al. (2009), the following text provides a foundational overview of anomaly detection suitable for the state-of-the-art section of a thesis:"Anomaly detection refers to the problem of finding patterns in data that do not conform to expected behavior. These non-conforming patterns are often referred to as anomalies, outliers, discordant observations, exceptions, aberrations, surprises, peculiarities or contaminants in different application domains. The importance of anomaly detection is due to the fact that anomalies in data translate to significant (and often critical) actionable information in a wide variety of application domains. At an abstract level, an anomaly is defined as a pattern that does not conform to expected normal behavior. A straightforward anomaly detection approach, therefore, is to define a region representing normal behavior and declare any observation in the data which does not belong to this normal region as an anomaly."+4For consistency analysis involving statistical comparisons between observed and expected values, the survey highlights the use of the χ^2 statistic:"The value of the χ^2 statistic is determined as:

$$\chi^2 = \sum_{i=1}^{n} \frac{(X_i - E_i)^2}{E_i} \tag{2.8}$$

where X_i is the observed value of the i_{th} variable, E_i is the expected value of the i_{th} variable (obtained from the training data) and n [is the number of variables]."

2.7.3 Bayesian Approaches

[19]Bell et al. First-principles Markov Chain Monte Carlo sampling is used to investigate uncertainty quantification and uncertainty propagation in parameters describing hydrogen kinetics. Rather than finding the best fit to kinetic parameters, our goal is to characterize the distribution of parameters that are consistent with a given set of experimental data. The resulting distribution identifies strong positive and negative correlations and several non-Gaussian characteristics. These results indicate the degree to which parameters consistent with the target experiments constrain predicted behavior in different combustion regimes.

[20]Braman et al. (2013) utilize a Bayesian framework to investigate the capabilities of syngas combustion models through the quantification of uncertainties. This framework, given a set of experimental data, "allows for the calibration of model parameters, determination of uncertainty in those parameters, propagation of that uncertainty into simulations, as well as determination of model evidence from a set of candidate syngas models". The state of knowledge about a parameter value is represented by a probability density function (PDF), which is updated according to Bayes' theorem:+4

$$P_{post}(\theta|d) = \frac{P_{prior}(\theta)\pi(\theta; d)}{\int P_{prior}(\theta)\pi(\theta; d)d\theta} \tag{2.9}$$

where P_{prior} is the prior PDF, P_{post} is the posterior PDF, and $\pi(\theta; d)$ is the likelihood function. The likelihood function "quantifies the level of agreement between the model and the data for specific values of the parameters". For model comparison, the authors employ the evidence function, which "measures the consistency of the model and the data considering the entire parameter space":+2

$$\pi_{evid}(M_i; d) = \int P_{prior}(\theta_i|M_i)\pi(\theta_i, M_i; d)d\theta_i \tag{2.10}$$

The study evaluates this approach with a "particular focus on kinetics parameter calibration and evidence-based chemistry model comparison" using laminar flame speed measurements from high pressure experiments.

[21]A novel toolchain is developed for Bayesian inference and uncertainty quantification (UQ) studies in practical hydrogen-enriched and lean-premixed combustion systems. The methodology implements an integrated chemical reactor network (CRN), non-intrusive polynomial chaos expansion using the point collocation method (NIPCE-PCM), and the Markov Chain Monte Carlo (MCMC) method. This framework is designed to characterize epistemic uncertainties in the modelling process, such as heat transfer, and to quantify the variability of input parameters through a two-step Bayesian calibration to maximize the agreement between model predictions and measurements.+4The stochastic process for quantities of interest (QoI), $\hat{y}$, is represented using a polynomial chaos expansion (PCE) defined as:

$$\hat{y} = \sum_{i=1}^{p} \alpha_i \Psi_i(\theta) \tag{2.11}$$

where θ is the vector of uncertain input parameters and $\Psi_i(\theta)$ represents multivariate polynomials. To reconcile experiments with the model, Bayes' theorem is implemented to transform a prior probability distribution, $P(\theta)$, for unknown parameters into a posterior distribution, $P(\theta|y)$:

$$P(\theta|y) = \frac{\mathcal{L}(\theta)P(\theta)}{\int \mathcal{L}(\theta)P(\theta)d\theta} \tag{2.12}$$

where $P(y)$ is experimental data and $\mathcal{L}(\theta) = P(y|\theta)$ represents a likelihood function.

2.7.4 MUM-PCE

In MUM-PCE, the uncertainty in the rate parameters of an as-compiled model is first quantified using stochastic spectral expansion and polynomial chaos techniques. The model is then constrained against global combustion experiments through solution mapping and least-squares optimization, combined with a consistency-screening procedure to identify incompatible experimental targets. The uncertainty of the constrained (posterior) model is subsequently calculated using an inverse spectral technique and propagated to a range of combustion conditions. [22]

2.7.5 B2BDC

Bound-to-Bound Data Collaboration (B2BDC) provides a natural framework for addressing both forward and inverse uncertainty quantification problems, in which QOI models are constrained by related experimental observations with interval uncertainty. B2BDC models uncertainty deterministically and the notion of validity is encapsulated in the consistency measure, and a collection of models, experimental observations, and parameter bounds determines a feasible region in the parameter space. If the feasible region is nonempty, the dataset is said to be consistent, whereas an empty feasible set indicates that models and data are fundamentally at odds. B2BDC is used to determine the consistency between a numerical model and an experimental dataset and to achieve reduction of the uncertainty, via inference from the data [23] [24]. A MATLAB implementation of B2BDC is provided by [12].

2.8 TUMKIN Implementations

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2.8.0.1 Bayesian History Matching

Implausibility metric

2.8.0.2 B2BDC Scalar Consistency Measure

Bound-to-Bound Data Collaboration (B2BDC) **Consistency Analysis for Massively Inconsistent Datasets in I**

Bound-to-Bound Data Collaboration (B2BDC) provides a natural framework for addressing both forward and inverse uncertainty quantification problems, in which quantity-of-interest (QOI) models are constrained by related experimental observations with interval uncertainty.

In this approach, uncertainty is modeled deterministically and the notion of validity is encapsulated in the consistency measure.

In B2BDC, models, experimental observations, and parameter bounds are taken as "tentatively entertained" and are combined into what is termed a dataset. A dataset determines a feasible region in the parameter space, defined as the set of parameter vectors for which models and data are in complete agreement within prescribed uncertainty bounds. If the feasible region is nonempty, the dataset is consistent, whereas if the feasible set is empty, then no parameter vector can lead to agreement between the models and the data.

A quantitative measure of agreement is provided by the scalar consistency measure, which characterizes this region by computing the maximal uniform constraint tightening associated with the dataset. The consistency measure is defined through an optimization problem that determines the largest simultaneous tightening of all experimental bounds for which a feasible parameter vector still exists. If the consistency measure is positive, all observation bounds can be simultaneously tightened without eliminating the feasible set. If the consistency measure is negative, the observation bounds are too restrictive and must be expanded in order for a feasible set to exist.

The B2BDC framework casts the problem of model validation in an optimization setting, where uncertainties are represented by intervals. Within a given physical model, parameters are constrained by the combination of prior knowledge and uncertainties in experimental data. A collection of such constraints constitutes a dataset and determines a feasible region in the parameter space. Assertions expressing additional knowledge, such as relationships among parameters or quantities of interest, can be incorporated as additional constraints.

B2BDC has been applied in several settings, including combustion science, atmospheric chemistry, quantum chemistry, system biology, and engineering design. Comparisons with Bayesian calibration approaches have shown that the two methods are similar in spirit and can yield overlapping predictions, while B2BDC provides conservative specifications and predictions when uncertainty information is limited.

In applied combustion modeling, B2BDC is used within a structured uncertainty-quantification workflow that includes quantification of experimental uncertainty, development and refinement of the physics model, identification of uncertain and sensitive parameters with prior bounds, sampling of the uncertain parameter space, and training of surrogate models. The final step, also referred to as the inverse problem, is performed by applying the Bound-to-Bound Data Collaboration approach to determine the consistency between a numerical model and an experimental dataset.

When datasets are inconsistent, B2BDC provides diagnostic tools to identify the sources of disagreement. Sensitivities of the consistency measure to perturbations in model-data constraints and parameter bounds are used to rank the degree to which individual constraints contribute to inconsistency. For datasets with numerous sources of inconsistency, an extension known as the vector consistency measure seeks the minimal number of independent constraint relaxations required to restore consistency.

Overall, B2BDC constitutes an optimization-based approach to model validation and uncertainty quantification, in which a model is considered predictive when its outcomes are accompanied by rigorously determined uncertainty bounds, and increased predictivity corresponds to more narrowly bounded interval estimations.

2.8.0.3 MUMPCE

Reliable simulations of reacting flow systems require a well-characterized, detailed chemical model as a foundation; however, the inherent uncertainties in the rate data leave a model characterized by a kinetic rate parameter space which will be persistently finite in its size. Without a careful analysis of how this uncertainty space propagates into the model predictions, those predictions can at best be trusted only semi-quantitatively. To address this issue, the Method of Uncertainty Minimization using Polynomial Chaos Expansions (MUM-PCE) is proposed to quantify and constrain these uncertainties.

In this method, the uncertainty in the rate parameters of the as-compiled model is quantified. The model is then subjected to a rigorous mathematical analysis by constraining the rate coefficients against the combustion data, as well as a consistency-screening process. Lastly, the uncertainty of the constrained model is calculated using an inverse spectral technique, and then propagated into a range of simulation conditions. The as-compiled model represents the prior model, while models constrained by global experiments are posterior models.

Uncertainty quantification in MUM-PCE is based on the stochastic spectral expansion method. The solution of a general dynamical model is expressed as a function of a vector of independent, identically distributed random variables,

$$u = u(\mathbf{y}, t, \boldsymbol{\xi}) \tag{2.13}$$

and expanded in a polynomial chaos basis. The expansion coefficients are obtained by projection using the expectation operator and orthogonality of the basis functions.

$$u(\mathbf{y}, t, \xi) = \sum_{i=0}^{\infty} u_i(\mathbf{y}, t) H_i(\xi) \tag{2.14}$$

To efficiently connect uncertainty quantification with chemical kinetic modeling, MUMPCE combines stochastic spectral expansion with solution mapping. Each uncertain rate parameter is normalized into a factorial variable,

$$x_i = \frac{\ln(k_i/k_{i,0})}{\ln f_i} \tag{2.15}$$

so that its nominal value is zero and its uncertainty is bounded between -1 and 1. Model responses are then expressed as a polynomial expansion in these variables,

$$g_r(\mathbf{x}) = g_{r,0} + \sum_{i=1}^{N} a_i x_i + \sum_{i=1}^{N} \sum_{j \geq i}^{N} b_{ij} x_i x_j + \cdots \tag{2.16}$$

with coefficients determined from sensitivity analysis or regression against computational experiments.

Uncertainty propagation is achieved by treating the factorial variables as random variables, which may be either uniformly or normally distributed. The uncertainty in a model prediction depends on the variance of the rate parameter but not strongly on the exact form of its distribution. For a sufficiently large number of active parameters, the resulting prediction uncertainty approaches a normal distribution by the Central Limit Theorem.

Model constraining is formulated as a least-squares optimization problem in which the difference between simulated responses and experimental observations is minimized subject to uncertainty bounds.

$$\Phi(\mathbf{x}^*) = \min_{\mathbf{x}} \sum_{r=1}^{n} \left(\frac{g_r(\mathbf{x}) - g_r^{\text{obs}}}{\sigma_r^{\text{obs}}} \right)^2 \tag{2.17}$$

A consistency analysis is introduced to evaluate whether experimental targets are internally consistent with the constrained model, using normalized scalar products and weighted residual measures to iteratively remove inconsistent targets.

Finally, uncertainty minimization is performed by interpreting the objective function as a joint probability density function for the rate parameters and deriving a covariance matrix for the posterior model. The reduction in uncertainty arises from strong coupling among rate parameters as the model becomes constrained by the experimental targets. Fundamental limitations remain due to the size, accuracy, and information content of available combustion data sets.

3 Methodology

Source code can be found at: https://github.com/barkinguney/consistency-analysis

3.1 Experimental Shock Tube IDT Uncertainty

Every dataset in TUMKIN needs to contain appropriate uncertainties. If published sources don't report uncertainties for experimental data, uncertainties are estimated using the algorihm proposed by **UQ_barkin**

The approach is aligned with the Guide to the Expression of Uncertainty in Measurement (GUM) [17], Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results [18]. The base principle follows from [17] and [18]: "the result of a measurement is only an approximation or estimate of the value of the specific quantity subject to measurement, that is, the measurand, y, and thus the result is complete only when accompanied by a quantitative statement of its uncertainty".

The reporting uncertainty is the expanded uncertainty U together with the coverage factor k used to obtain it, or report combined uncertainty u_c, which represents the estimated standard deviation of the result.

The reporting a measurement result and its uncertainty, include the information of all components of standard uncertainty, which may be grouped into two categories according to the method used to estimate their numerical values: type A, those which are evaluated by statistical methods, i.e. the corresponding component of uncertainty arising from a random effect; type B, those which are evaluated by means other than the statistical analysis of series of observations. In our analysis we will use a simple correspondence between the classification of uncertainty components into categories A and B and the commonly used classification of uncertainty components as "random", denoted $\sigma_r(y)$, and "systematic," denoted $\sigma_\delta(y)$. The combined standard uncertainty is calculated as the square root of the sum of the variances from systematic and random components

$$u_c(y) = \sqrt{\sigma_\delta(y)^2 + \sigma_r(y)^2} \tag{3.1}$$

$\sigma_r(y)^2$ is estimated combining the individual standard uncertainties σ_i of the uncertainty components applying the law of uncertainty propagation.

The *systematic error* $\sigma_\delta(y)$ captures device- and condition-specific biases in τ_{ign} measurements in shock tubes. As these conditions are typically modelled using a 0-D mathematical framework that accounts solely for chemical processes behind of shock wave, i.e. an ideal case. Alternative IDT modeling approaches are outside the present scope. To evaluate $\sigma_\delta(y)$ we developed a statistical procedure that quantifies the departure from this idealization. Concretely: (i) we identify the dominant facility/operating-condition factors provoke non-idealities behind of shock wave; (ii) construct estimators for each factor's contribution to $\sigma_\delta(y)$; (iii) define a quasi-ideal reference regime—a subset of facilities/conditions that meet strict criteria—and assign to this reference a baseline systematic error of 5%; (v) establish the quantitative metric of deviation from reference case; and (iv) for factors outside the reference, we compute incremental penalties and aggregate these contributions to obtain $\sigma_\delta(y)$. This construction makes explicit the assumption that a small, irreducible systematic component (≈5%) exists even under best-case conditions, with additional bias driven by quantifiable departures from the ideal case. Consequently, the systematic error, we will evaluate, can be classified as an "offset or zero setting error".

Because rigorous, quantitative assessments of systematic error in τ_{ign} data remain scarce, the primary focus of this work is on addressing this gap; details follow in the next sections.

The final uncertainty is expressed as a coverage interval, where U is the expanded uncertainty obtained by multiplying $u_c(y)$ by a coverage factor, k. Thus

$$U = ku_c(y) \tag{3.2}$$

and it is confidently believed that $y-U \leq Y \leq y+U$, which is commonly written as $Y = y \pm U$. To be consistent with current international practice, the value of k to be used for calculating U is, by convention, $k = 2$.

3.1.1 Systematic Uncertainty

Guided by non-ideality phenomena in shock tubes, 14 input/operating factors (IFs) are selected, Figure 3.1, that determine proximity to the 0-D "ideal case", i.e., conditions under which τ_{ign} can be described with the 0-D mathematical model. A quasi-ideal reference is defined by fixing each factor to its target metric $\boldsymbol{\mu}_i^0$ Figure 3.1 and assign a baseline systematic error $\delta^0 = 5\%$ at 95% confidence. This baseline represents the irreducible risk under best-case conditions; deviations from $\boldsymbol{\mu}_i^0$ are treated separately.

Category	#	Impacting Factor (IF)	Target metric, $\boldsymbol{\mu}_i^0$
Test equipment	1	Driver length	≥ 3.0 m
	2	Driven length	≥ 8.0 m
	3	Driven inner diameter	≥ 10.0 cm
	4	Tailoring	yes
	5	Inserts	yes
	6	CRV	yes
Operating conditions	7	Reflected-shock Temperature, T_5	1000 - 1600 K
	8	Reflected-shock Pressure, P_5	5 - 20 atm
	9	Valid test time, Δt_{valid}*	50 - 500 µs
	10	Dilution (Ar/He/…)**	yes
	11	Impurities	no
	12	Measurement location	≤ 2cm (from the end of the tube)
	13	Fuel-oxidizer equivalence ratio	≥ 0.3
	14	Pre-shock pressures, P_1	≥ 10 Torr

*Δt_{valid} time window before contact-surface/expansion-wave arrival or significant boundary-layer growth invalidates 0-D assumptions.
** "Yes" indicates purposeful dilution to suppress non-ideal gas dynamics and heat-release effects (typical practice is high inert fraction; specify per system if needed).

Figure 3.1: Impacting factors (IF) and target metrics for the "ideal case" $\delta_0 = 5\%$ (95% confidence).

GUM-consistent recipe to turn Type-B systematic term δ into a combined uncertainty $u_c(y)$, eq. (1), with Type-A random. So, calculated in this section δ tis systematic expanded relative uncertainty will be later transformed in the systematic standard uncertainty.

We calibrate the systematic error against a quasi-ideal reference through departures of IFs from its reference target. Assuming the independence of all other IFs these departures are mapped to incremental penalties via a smooth "undesirability" transform, and aggregated as

$$\delta = \delta^0 + \sum_{i=1}^{N} \delta_i = \delta^0(1 + \sum_{i=1}^{N} \xi_i), \quad \delta_i = \delta^0 \xi_i \ , \quad i = 1, \ldots, N, \tag{3.3}$$

where N is the number of IFs, and $\xi_i \in [0, 1)$ quantifies the penalty induced by the deviation of μ_i from the ideal μ_i^0.

To define and ξ_i, we use a two-sided, smaller-is-better modification of Harrington's desirability concept [48] [49] retaining low sensitivity near the ideal point and saturation for large deviations (Fig. 4):

$$\xi_i = 1 - e^{-(x_i^2)} \tag{3.4}$$

$$x_i = (\mu_i^0 - \mu_i)/\mu_i^0 \; . \tag{3.5}$$

For binary IFs (e.g., tailoring, inserts, CRV, dilution, impurities), we map an unfavorable setting to the conservative lower edge of the "satisfactory" band, grey color in Fig. 4, i.e. $\xi_i = e^{-1} \cong 0.37$, if "no" and to $\xi_i = 0$ if "yes" for $\mathrm{IF}_{4,5,6,10,13}$. For IF_{11} it is otherwise.

As the test time (IF_9) is strongly correlated with T_5 and P_5, we retain IF_9 in the systematic term and exclude T_5 and P_5 from δ (their uncertainties enter the random part later). We also exclude IF_{14} (P_1) due to a lack of quantitative evidence.

3.1.2 Random Uncertainty

The measured value should be reported along with an estimate of the total combined standard uncertainty, sum of systematic error and random error. Formally, the rigorous approach to evaluate the random error of a measurement would require the repetition of the same measurement for a significant number of times, what is non-realistic to perform in praxis or from the rules for differentiating the measurement model describing the indirect measured target. In this case, the measured data set is approximated with the function of several parameters and the standard deviation of a measurand is calculated as the positive square root of the combined variance, $u_c^2(y)$, assuming independence of input parameters [17]:

$$u_c^2(y) = \sum_{i=1}^{N} \left(\frac{\partial f}{\partial z_i} \right)^2 u^2(z_i) + 2 \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{\partial f}{\partial z_i} \frac{\partial f}{\partial z_j} u(z_i) u(z_j) r(z_i, z_j), \tag{3.6}$$

where z_i are parameters of the function f; $u^2(z_i)$ is the standard uncertainty of parameters; $r(z_i, z_j)$ is the estimated correlation coefficient.

To describe the set of measured IDT data we used the function proposed by A. Lifshitz and co-authors for data processing [19]:

$$\tau_{ign} = f(z_i, A, B, C, D) = A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D \, . \tag{3.7}$$

Here $z_1 = T_5$ is the temperature behind the shock wave, $z_2 = P_5$ is the pressure behind the shock wave, and $z_3 = \varphi$ stoichiometric ratio, fuel/oxygen. A, B, C, D – parameters of approximation.

We apply eqs. (4) and (5) under assumptions A, B, C, D, T_5, P_5, and φ are not correlated parameters and terms with parameter correlations can be neglected.

In this case

$$\begin{aligned}\sigma_i^2(\tau_{ign}) = &\left(z_2^C z_3^D A \exp\left(\frac{B}{z_1}\right) \cdot \left(-\frac{B}{z_1^2}\right)\right)^2 \cdot u^2(z_1) + \left(CA \exp\left(\frac{B}{z_1}\right) z_2^{C-1} z_3^D\right)^2 \cdot u^2(z_2) \\ &+ \left(DA \exp\left(\frac{B}{z_1}\right) z_2^C z_3^{D-1}\right)^2 \cdot u^2(z_3) + \left(\exp\left(\frac{B}{z_1}\right) z_2^C z_3^D\right)^2 \cdot u^2(A) \\ &+ \left(\frac{1}{z_1} A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D\right)^2 \cdot u^2(B) + \left(A \exp\left(\frac{B}{z_1}\right) z_2^C \ln(z_2) z_3^D\right)^2 \cdot u^2(C) \\ &+ \left(A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D \ln(z_3)\right)^2 \cdot u^2(D) \end{aligned} \tag{3.8}$$

The standard uncertainty of input parameters $u^2(z_i)$, i.e. manufacture's specifications, calibration, and other certificates should be reported in a paper. If it is not so it can be evaluated from the available information, analogy, etc.

Weighted Least Squares (WLS) regression is used to calculate best A,B,C,D parameters and their uncertainties $u^2(A), u^2(B), u^2(C), u^2(D)$. If all data points in a data set have very similar values for any of the variables T5, P5, ϕ the fit is multicollinear. This makes the calculated uncertainties meaningless. This is especially common for ϕ since an IDT dataset usually contains measurements at constant ϕ. To prevent this, if a variable has constant value across the dataset, Equation 3.7 is modified to set the corresponding parameter B,C or D set to 1. In the case of constant ϕ, Equation 3.7 becomes:

$$\tau_{ign} = f(z_i, A, B, C) = A \exp\left(\frac{B}{z_1}\right) z_2^C z_3. \tag{3.9}$$

Equation 3.8, that propagates parameter uncertainties to τ_{ign} uncertainty is also recalculated with the new parameter set A,B,C.

Normally, T_5, P_5, and φ are reported in a paper with uncertainty interval given as

$$Z = z \pm U(z). \tag{3.10}$$

Assuming normal distribution of x_i and 95% of confidence interval one can interpret U_i as uncertainty interval

$$U(z_i) = 2\sqrt{u_c^2(z_i)} \tag{3.11}$$

and the standard uncertainty of parameters can be calculated as

$$u^2(z_i) = \frac{U^2(z_i)}{4}. \tag{3.12}$$

The standard uncertainty of a set of IDT data can be calculated from the sum of systematic and random uncertainties, Table 2 and eq. (6)

$$u_{ci}(\tau_{ign}) = \sqrt{\sigma_{\delta_i}^2 + \sigma_i^2(\tau_{ign})} \tag{3.13}$$

$\sigma_{\delta_i}^2$ is the standard systematic deviation. As we interpret δ as normal, 95% expanded, we obtain:

$$\sigma_{\delta_i}^2 = \left[\tau_{ign} * \left(\frac{\delta_i/2}{100}\right)\right]^2 \tag{3.14}$$

The final uncertainty interval of the measured set will be calculated as (in the case of normal distribution of x_i and 95% of confidence interval):

$$Ui(\tau_{ign}) = 2(u_{ci}(\tau_{ign})) \tag{3.15}$$

Finally, data accepted for the further analysis and model testing, fitting and optimization are reported as:

$$Y^{exp} = \tau_{ign}^m \pm U(\tau_{ign}) \tag{3.16}$$

where τ_{ign}^m is measured value of the IDT.

For the same experimental set from the study [50], we calculate now the random error applying described procedure. Uncertainty of input parameters were calculated with data reported in the paper of S. Saxena et al. [50]: reflected shock temperature and pressure uncertainties are ±1% and ±1.5% respectively; uncertainty in test gas composition before shock heating was estimated to be less than 0.5%.

3.2 Arrhenius Parameters Uncertainty

Different literature sources report different values for the Arrhenius parameters or rate coefficients for the same reaction. Therefore, it is important to take all this information into account when analysing and optimizing reaction mechanisms. This package collects kinetic data from a database and assigns a single uncertainty factor f to each reaction, using the collected data.

3.2.1 Data Collection

To evaluate the uncertainty of a specific reaction, all instances of that reaction across all loaded kinetic libraries in the RMG (Reaction Mechanism Generator) database are identified. Reactant and product species are determined by comparing the reaction equation string to known species. SMILES representations of the species are used to search RMG database libraries for matching sources. For every match, rate coefficients $k(T)$ are evaluated across a grid of the valid temperature range of each reaction. A downside of RMG database is very few sources include a valid temperature range for the reported kinetic parameters. In that case a standardized temperature grid $T \in [300, 3100]\,\mathrm{K}$ is used to collect rate coefficient data. Collected kinetics data, and rate coefficients evaluated at fixed intervals within the temperature range are saved further processing. Currently RMG kinetics types of Arrhenius, MultiArrhenius, ThirdBody are supported. For future development, support for PDepArrhenius should be added. Currently efficencies of ThirdBody reactions are not considered.

3.2.2 Uncertainty Factor

The modified Arrhenius equation is expressed as:

$$k(T) = AT^n \exp\left(-\frac{E_a}{RT}\right) \tag{3.17}$$

To perform a robust statistical fit across multiple orders of magnitude, equation is transformed into logarithmic space.

$$\ln k(T) = \ln A + n \ln T - \frac{E_a}{RT} \tag{3.18}$$

In standard Arrhenius fitting parameters (A,n,E_a) are usually highly correlated.

$$\ln k(T) = \beta_0 + n \ln\left(\frac{T}{T_{ref}}\right) - \frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right) \tag{3.19}$$

The rate expression is transformed into a log linear model centered around a reference temperature, which is geometric mean of temperature unless otherwise specified. The parameters

(β_0, n, E_a) are solved using a Nonlinear Least Squares approach, minimizing the residual sum of squares (RSS) between the library data and the model. The resulting covariance matrix in the reparameterized space, Σ_β, is then mapped back to the original $(\log_{10} A, n, E_a)$ space using the Jacobian transformation matrix J:

$$\Sigma_\theta = J \Sigma_\beta J^T \tag{3.20}$$

Diagonal entries of the covariance matrix of the least squares fit give parameter standard deviations $s\left(\log_{10}(A)\right), s(n), s(E_a)$, which are used to calculate lower and upper uncertainty bounds for rate coefficients.

$$k_{\text{low}}(T) = 10^{\log_{10}(A) - s(\log_{10} A)}\, T^{n - s(n)} \exp\left(-\frac{E_a + s(E_a)}{T}\right) \tag{3.21}$$

$$k_{\text{up}}(T) = 10^{\log_{10}(A) + s(\log_{10} A)}\, T^{n + s(n)} \exp\left(-\frac{E_a - s(E_a)}{T}\right) \tag{3.22}$$

The uncertainty parameter f is defined as:

$$f(T) = \log_{10}\left(\frac{k_{\text{up}}(T)}{k_0(T)}\right) = \log_{10}\left(\frac{k_0(T)}{k_{\text{low}}(T)}\right) \tag{3.23}$$

Where k_0 is the best fit of the rate coefficient, k_low and k_up are the lower and the upper bounds respectively [9]. This gives a temperature dependent uncertainty factor f(T). Temperature dependence of f(T) is usually low. To simplify further analysis, mean value of f(T) is selected as constant f. When rate coefficients need to be sampled in further anylysis, they are sampled from the interval $\left[k_0(T)/10^f(T), k_0(T) * 10^f(T)\right]$. In practice this can be achieved by multiplying the pre exponantial factor A witha multiplier sampled from $\left[(T)/10^f(T), (T) * 10^f(T)\right]$. Which preserves the shape of the k(T) curve and only scales it.

$$f_{final} = \max_T f(T) \tag{3.24}$$

This uncertainty factor provides a single, conservative metric to quantify uncertainty of the reaction rate coefficient across the entire temperature range.

3.3 Cantera IDT Simulations

Cantera is a 0D and 1D chemical kinetics simulation software. To simulate shock tube IDT, constant volume batch reactors are used. Ignition delay times to be simulated for a given operating conditions T5, P5 phi and mechanism. The exact measurement or simulation time of IDT vary based on IDT definition. Using the same IDT definition as reported by the physical experiment to calculate simulated IDT would yield the closest results to physical experiment. the The ignition types that can be handled by this IDT simulation code are a follow. Following the format described by RESPECTH shock tube IDT measurements, ignition type is comprised of attributes, target, type, amount and unit . Target can be pressure, temperature or concentration. Type can be “max”, “d/dt max”, “baseline max/min intercept from d/dt“, “concentration”, “relative concentration”, and “relative increase”. ”max” means IDT is at the maximum of the target property. “d/dt max” means IDT is at the maximum gradient of target property. “baseline max/min intercept from d/dt“ means the time when the max d/dt line intercepts the baseline value of target property. “concentration”: IDT is at the time target property reaches a specified concentration value. “relative concentration”: IDT is at the time when the concentration of the target property relative to its maximum concentration reach the specified value. “relative increase”:IDT is at the time when the value of the target property increases by a given relative amount compared to its value at t=0. Amount attribute is only used when type are “concentration”, “relative concentration” or “relative increase”. Unit attribute is only used when the type is “concentration”, “relative concentration” or “relative increase”. Unit should have the value “unitless” for relative types.

3.4 Consistency analysis

Consistency methods, that check for a given dataset of experimental ignition delay time measurements and a mechanism with parameter bounds, whether experiments and mechanism are consistent with each other.

Figure 3.2: Flowchart.

3.4.1 Sensitivity analysis

For a given QoI, the model responses are sensitive to only a small subset of parameters. These parameters, referred to as active parameters, dominate the response behavior, a phenomenon referred to as effect sparsity [25]. Employing a sensitivity analysis method to identify these active parameters and using only these active parameters in further analysis improves computational feasibility. In [12] active parameters are identified by sampling the prior uncertainty region using space-filling designs like Latin hypercube sampling [26] or Sobol quasi-random sequence [27], and evaluating the original model at those points. A linear regression model Sl is then fitted to approximate the QoI as a function of the parameters.

$$S_l = \sum_{i=1}^{n} c_i x_i + c_0 \tag{3.25}$$

S_l contains information about how much the model output changes when each parameter is varied around its nominal value. The resulting regression coefficients define impact factors, which estimate how much the model output changes when each parameter varies across its uncertainty range. This is similar to sensitivity, but it also considers how large the parameter uncertainties are. A parameter with low sensitivity can still be high impact if it has high uncertainty.

$$\mathrm{IF}_i = |c_i| \ (u_i - l_i) = |c_i| \ \Delta x_i \tag{3.26}$$

Active parameters are selected by screening the ranked impact factors with a preset threshold [12]. Depending on the use case sensitivity analysis or uncertainty analysis, S_l or $\mathrm{I}Fi$ is used to rank parameters.

3.4.2 B2BDC Integration

3.5 Data Sources

When TUMKIN is up and running, uncertainty quantification and consistency analysis codes described in this thesis will use the data available in TUMKIN. The following sources has been used in the mean time to develop these modules, and could be imported into TUMKIN if suitable. Kinetics database RESPECTH [2] collect IDT, laminar flame speed, and concentration measurement s for various fuels. They are reported in their xml format. For testing the modules developed in this thesis I have used the RESPECTH data for shock tube IDT data for ethylene, Syngas and Hydrogen. Currently Most comprehensive collection of reaction rates and Arrhenius parameters is the NIST Kinetics Database [20]. That contain many literature sources for every reaction. However, they don't allow access to their database from code, no API or anything. It is the most comprehensive source, however, must be accessed manually. Reaction Mechanism Generator (RMG) on the other hand also contains a kinetics database that is openly accessible. For this thesis Arrhenius parameter data s taken from RMG database. To algorithmically extract kinetics data from RMG for a given reaction, SMILES or INCHI representations of the products and reactants are necessary. So, a species database that map between the widely accepted SMILES, INCHI representations and Cantera species name, and composition is necessary. An example format can be seen in OPTIMA++. Species names for the same species can be different across mechanism files if they are consistent within the mechanism. This is however not suitable for TUMKIN database. So before ingesting Mechanism files into TUMKIN it should be validated that species in mechanism can unambiguously be found in the TUMKIN species data base. This validating step is not in the scope of this thesis however is necessary for TUMKIN.

4 Results and Discussion

In this thesis tools for Arrhenius paramter uncertainty, Experimental IDT uncetainty, sensitivity analysis have been developed and intergrated with B2BDC for consistency anylsis.

4.1 B2BDC Consistency

Joint Syngas/Hydrogen mechanism from [28] is used. Wang et. al report that among others, experimental IDT data by TAMU[29] show good data consistency with their model. [28]. To test the integration of the B2BDC MATLAB toolbox with this thesis work, same IDT data will be used. 4.34.44.64.6 are generated by the MATLAB toolbox provided by [12]

4.1.1 Good Data

Cantera IDT simulations are used to generate training data. Quadratic surrogate models for each data point are built from the training data. The B2BDC consistency check results in a scalar consistency measure interval of $[0.6118, 0.7261]$. Since the lower bound is positive, this demonstrates that the dataset and the model are consistent, and a feasible set of model parameters exists.

4.1.2 Good Data + 1-2 Bad data

Cantera IDT simulations are used to generate training data. Quadratic surrogate models for each data point are built from the training data. The B2BDC consistency check results in a scalar consistency measure interval of $[-3.0078, -0.5989]$. This proves dataset inconsistency, as the conservative bound (outer bound) of the scalar consistency measure is negative [12]. Figure 4.6 and Figure 4.7 rank the observation and constraint bounds to which the consistency is most sensitive. As expected, the artificially added "bad" data points (12) and (13) are the largest suspected causes of the inconsistency. In the next step, these data points are removed

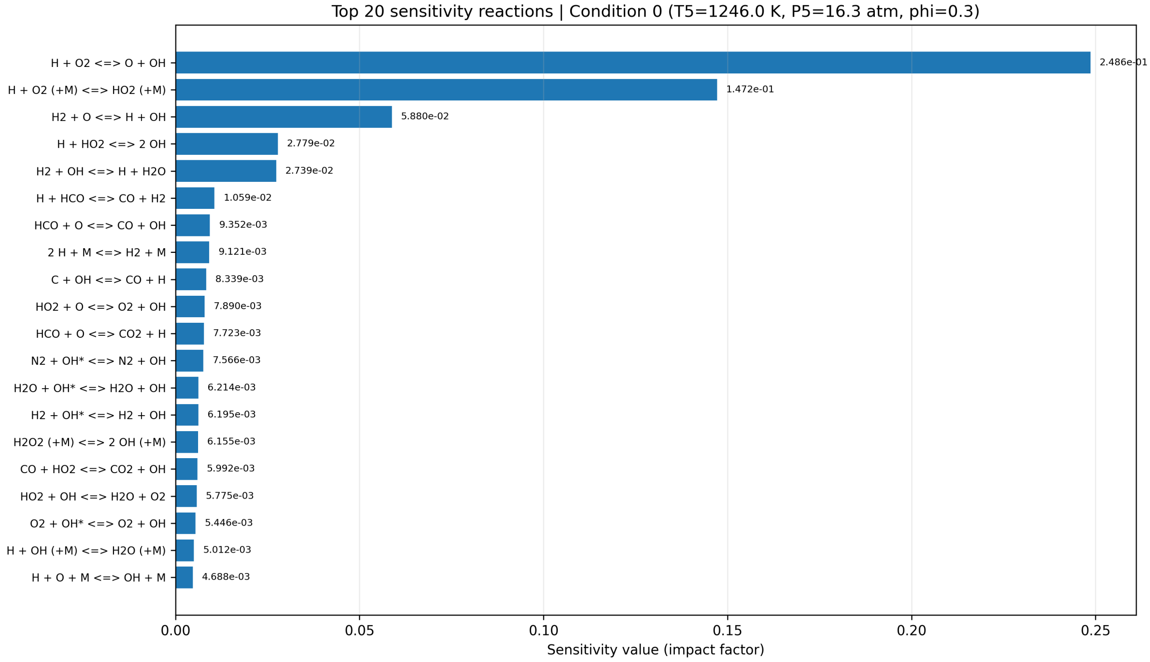


Figure 4.1: Sensitivity of IDT to the reaction rate coefficients at a single operating condition. Condition from [29]. $H_2/O_2/Ar$ mixture, $\phi = 0.3$, $P = 16.3$ atm

from the dataset. The scalar consistency measure now yields $[0.6118, 0.7261]$. Since the lower bound is now positive, this demonstrates that the dataset and the model are consistent, and a feasible set of model parameters exists. Because the dataset and model are now equivalent to those discussed in Section 4.1.1, the sensitivities of the scalar consistency measure's inner and outer bound solutions to the QoI and model parameter uncertainty bounds are also equivalent to Figure 4.3 and Figure 4.4.

4.2 Uncertainty of Shock Tube IDT Data

4.2.1 shen vs davidson

4.2.2 Syngas

for most reported sources shock tube setup specific information is not available in an easily accessable database. They are available in the original papers,however to extract them would

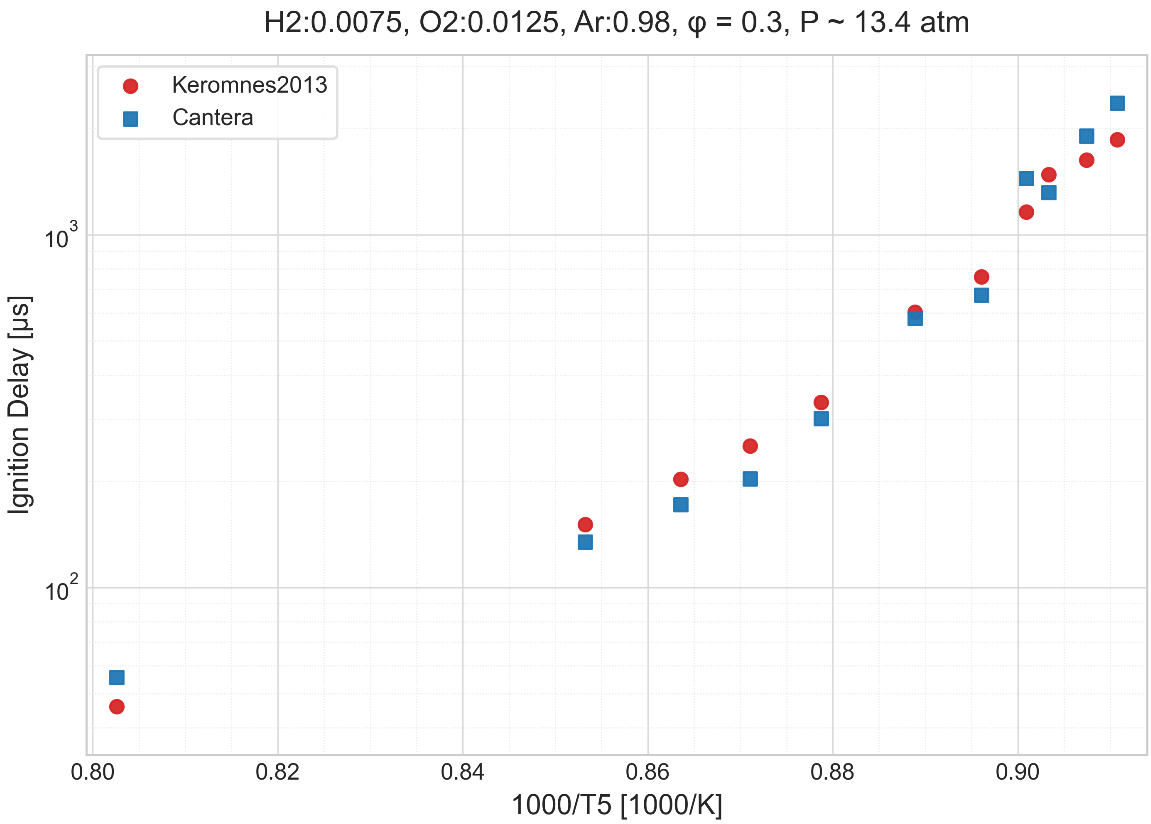


Figure 4.2: IDT measurements [29]. $H_2/O_2/Ar$ mixture, $\phi = 0.3$, $P \sim 13.4$ atm

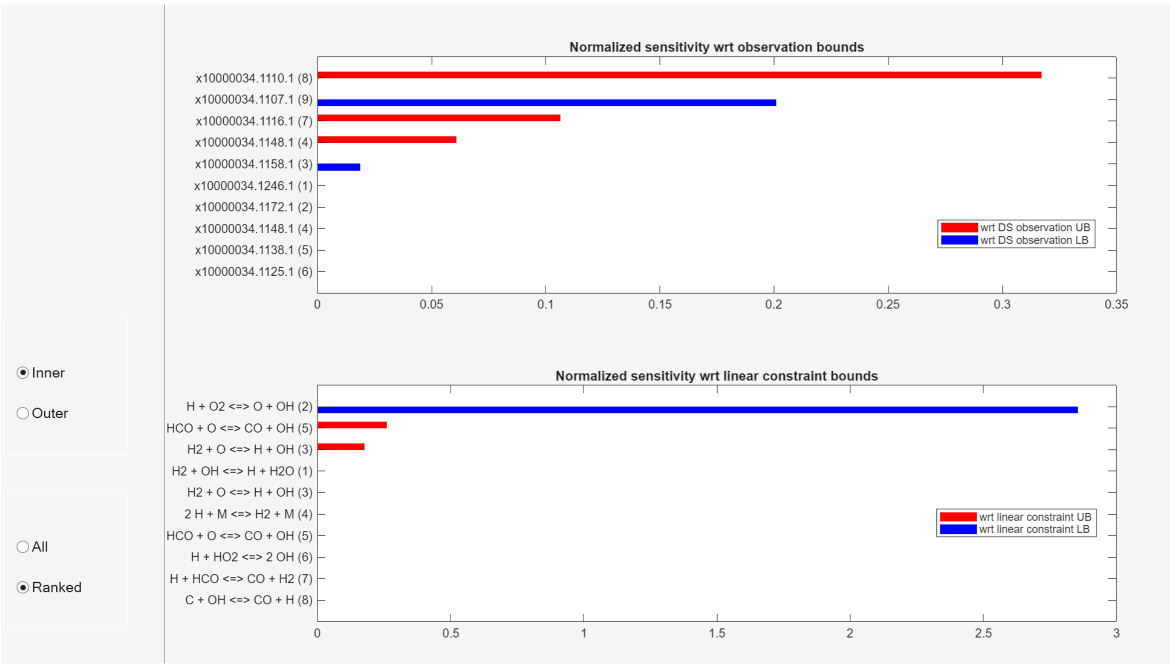


Figure 4.3: Ranked sensitivity of scalar consistency measure (SCM) inner-bound solutions with respect to the quantity of interest (QOI) and model parameter uncertainty bounds.

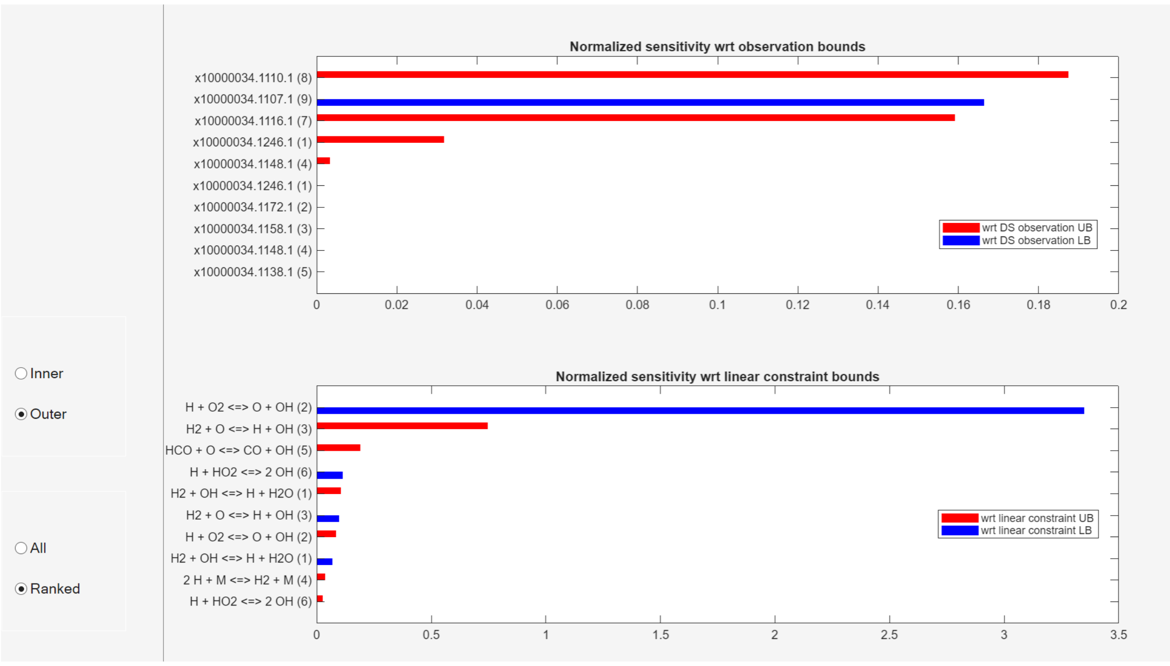


Figure 4.4: Ranked sensitivity of scalar consistency measure (SCM) outer-bound solutions with respect to the quantity of interest (QOI) and model parameter uncertainty bounds.

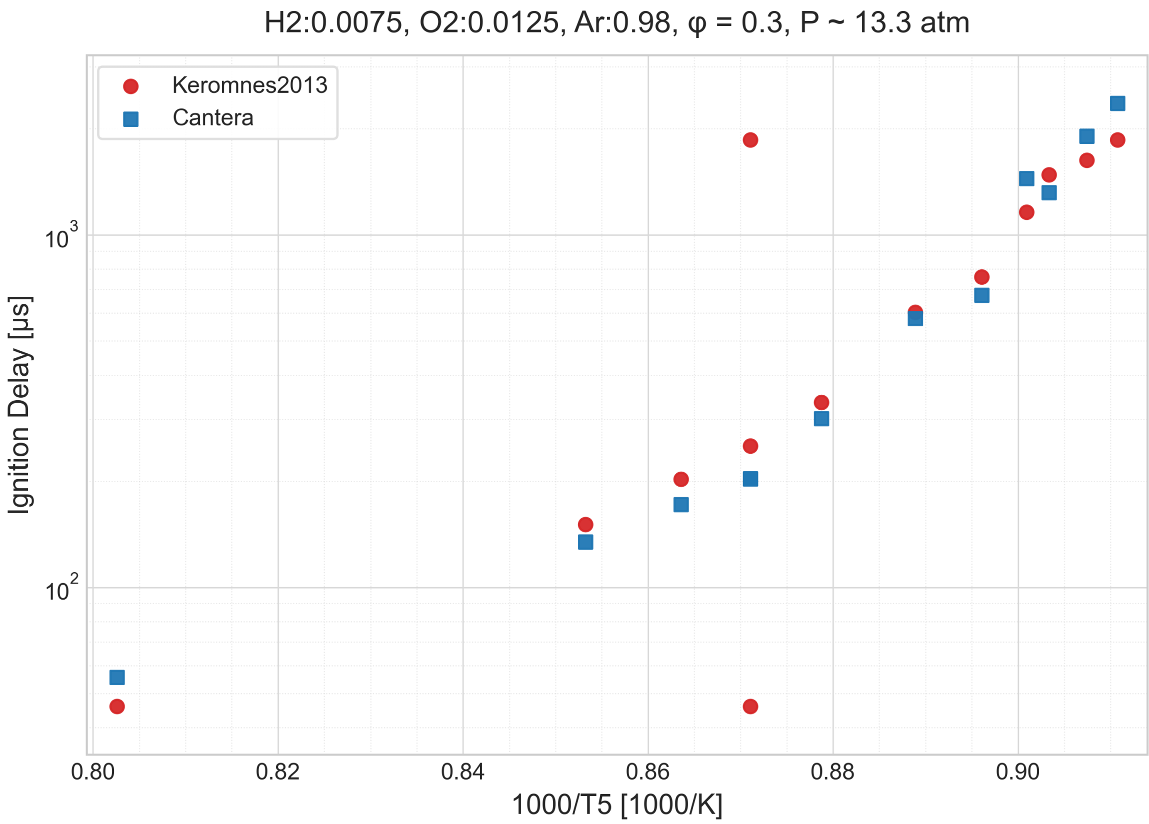


Figure 4.5: IDT measurements [29] + 2 artificial "bad" data points.
$H_2/O_2/Ar$ mixture, $\phi = 0.3$, $P \sim 13.3$ atm

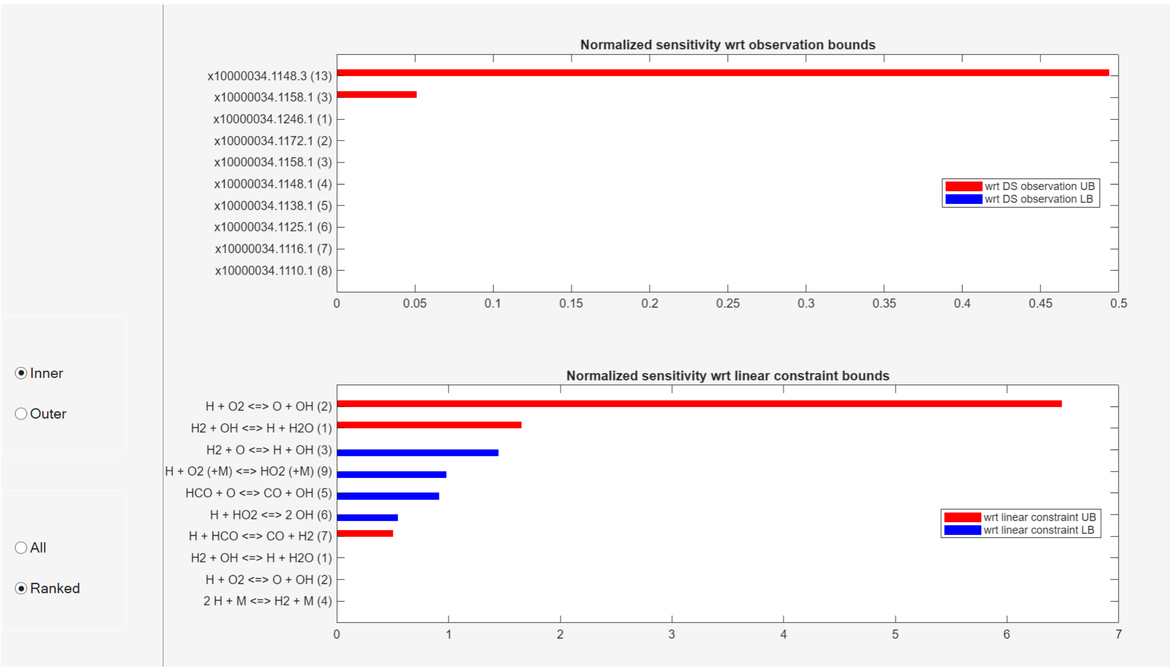


Figure 4.6: Ranked sensitivity of scalar consistency measure (SCM) inner-bound solutions with respect to the quantity of interest (QOI) and model parameter uncertainty bounds. Dataset includes inconsistent data.

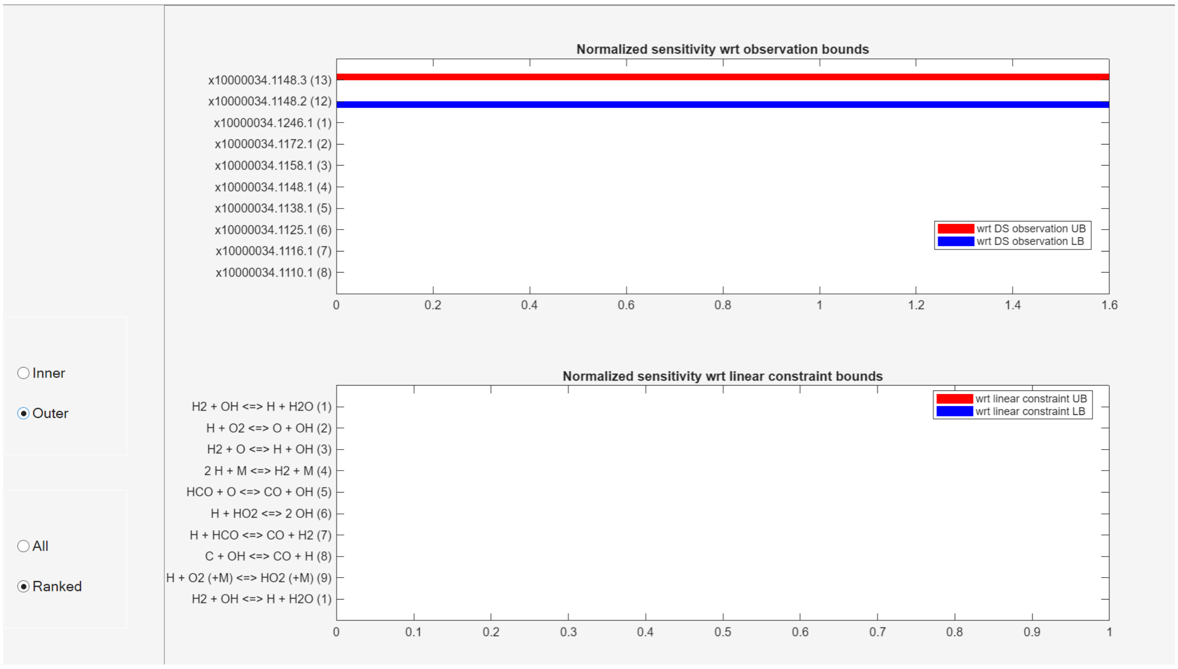


Figure 4.7: Ranked sensitivity of scalar consistency measure (SCM) outer-bound solutions with respect to the quantity of interest (QOI) and model parameter uncertainty bounds. Dataset includes inconsistent data.

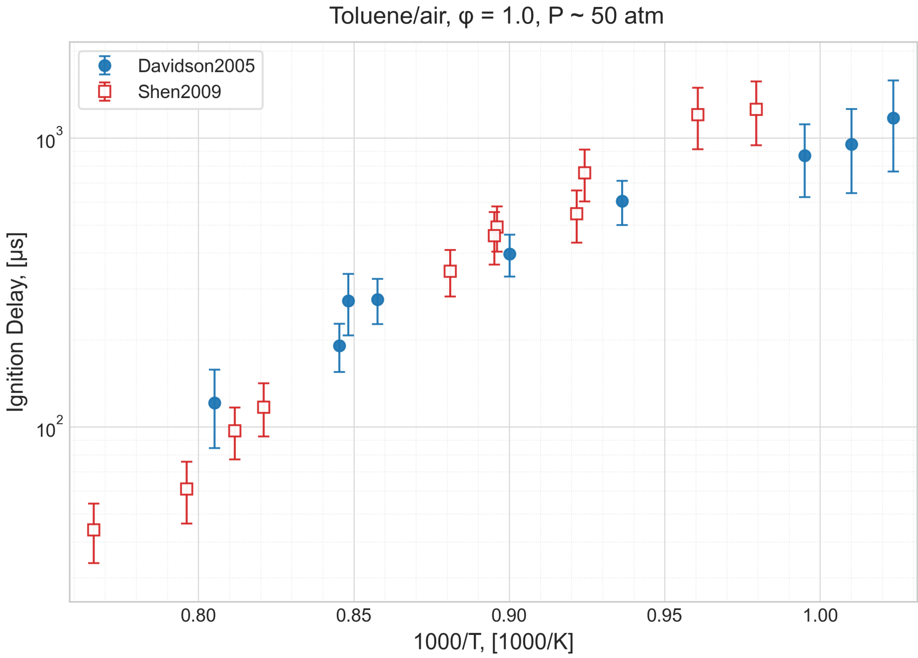


Figure 4.8: Comparison of calculated IDT uncertainties for a stoichiometric toluene/air mixture. IDT measurements are from two different facilities [30][31] over similar temperature range. P∼50 atm.

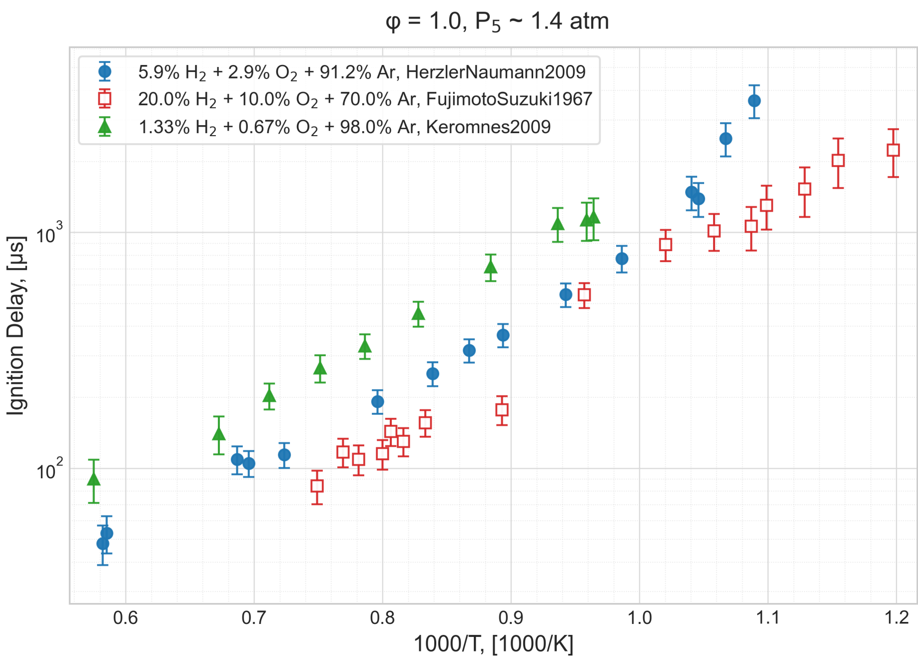


Figure 4.9: Comparison of calculated IDT uncertainties for $H_2/O_2/Ar$ mixtures. IDT measurements are from sources [32] [33] [29].$\phi = 1.0, P \sim 1.4 atm$.

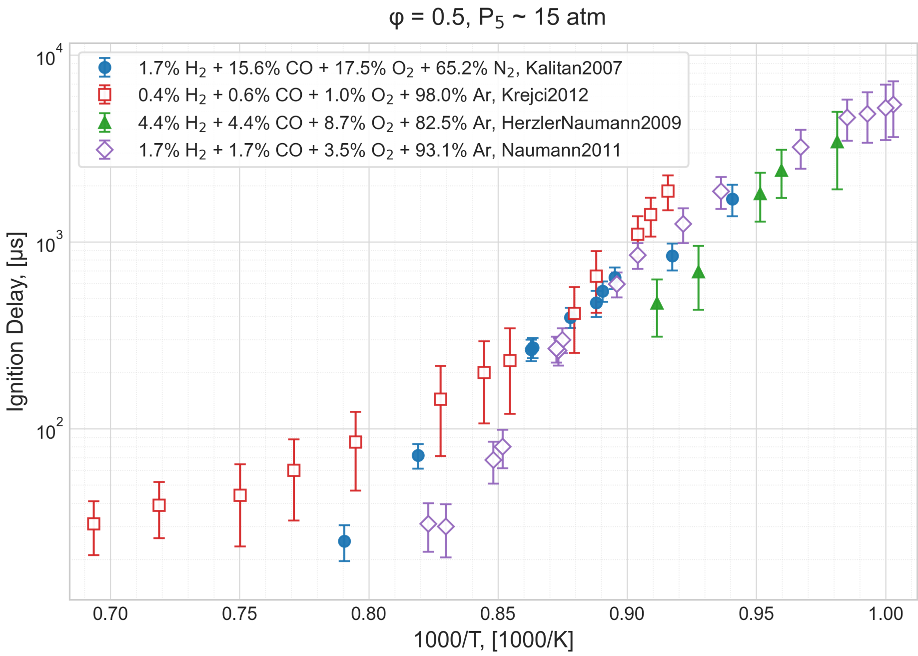


Figure 4.10: Comparison of calculated IDT uncertainties for diluted syngas/oxidizer mixtures. IDT measurements are from sources [34] [35] [32] [36]$\phi = 0.5, P \sim 15atm$.

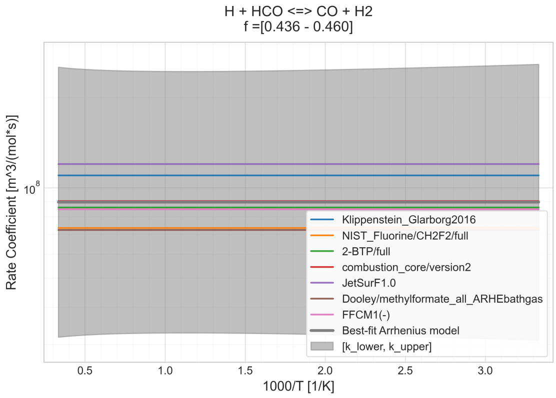
require manual entry, for programmatic processing with language model. This is necessary for an accurate estimation of experimental shock tube IDT uncertainties.

4.3 Reaction rate coefficient uncertainty

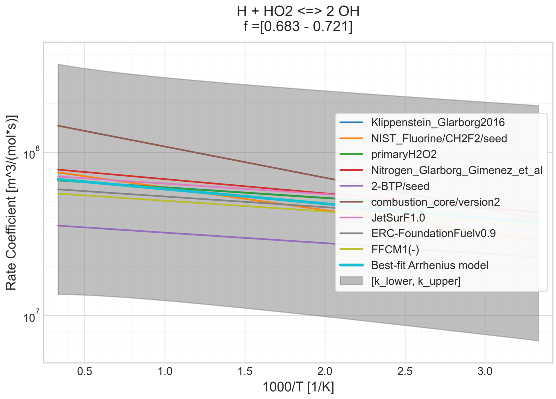
Uncertainty factors have been calculated using all available libraries in RMG database. If there are not enough sources in the database for a given reaction, uncertainty factor calculation cannot be performed.

4.4 Cantera Shock Tube IDT Simulations

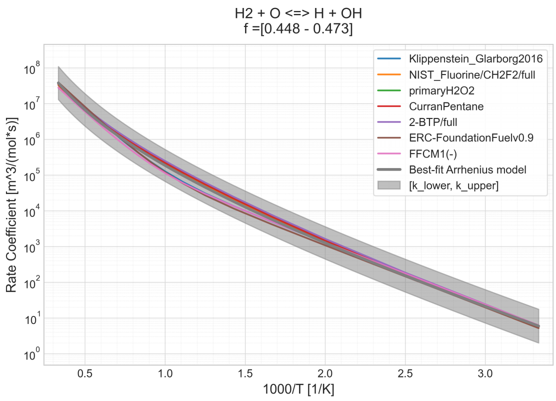
Current implementation of Cantera IDT simulation is tested with all available Syngas and Hydrogen IDT data from RESPETCH database. 4.12, 4.13, 4.14 show some example comparisons from that dataset.



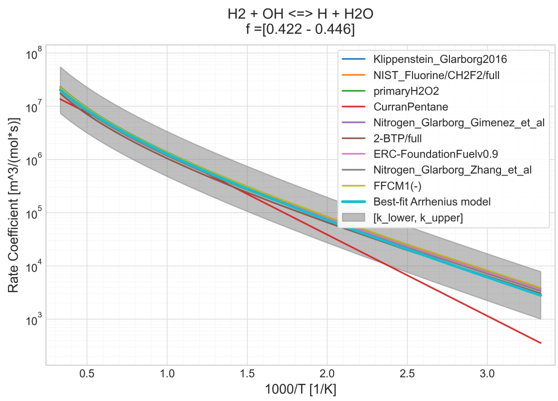
(a) Uncertainty factor for $H + HCO <=> CO + H_2$ [6]



(b) Uncertainty factor for $H + HO_2 <=> 2OH$ [6]



(c) Uncertainty factor for $H_2 + O <=> H + OH$ [6]



(d) Uncertainty factor for $H_2 + OH <=> H + H_2O$ [6]

Figure 4.11: Comparison of best fit reaction rate coefficients with source reaction rate coefficients [6]

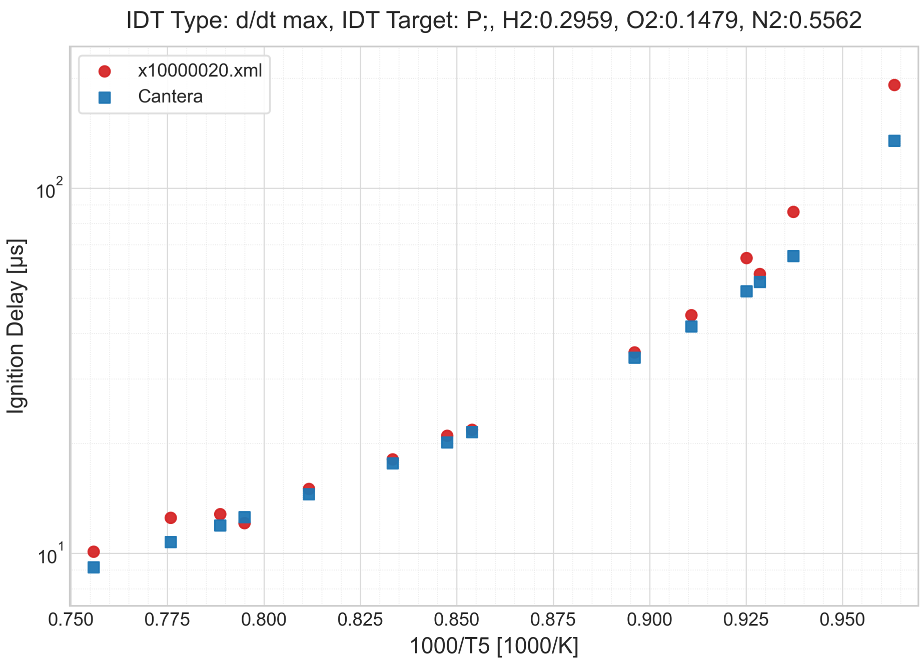


Figure 4.12: Comparison of Cantera IDT simulation with experimental measurement **good_cantera** IDT Type: dP/dt max.

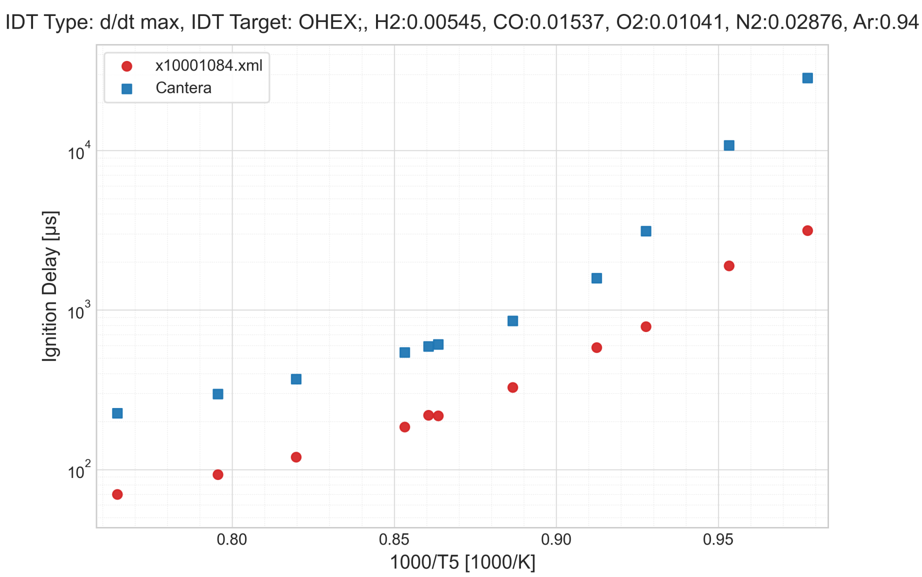


Figure 4.13: Comparison of Cantera IDT simulation with experimental measurement **good_cantera** IDT Type: d(OH^*)/dt max.

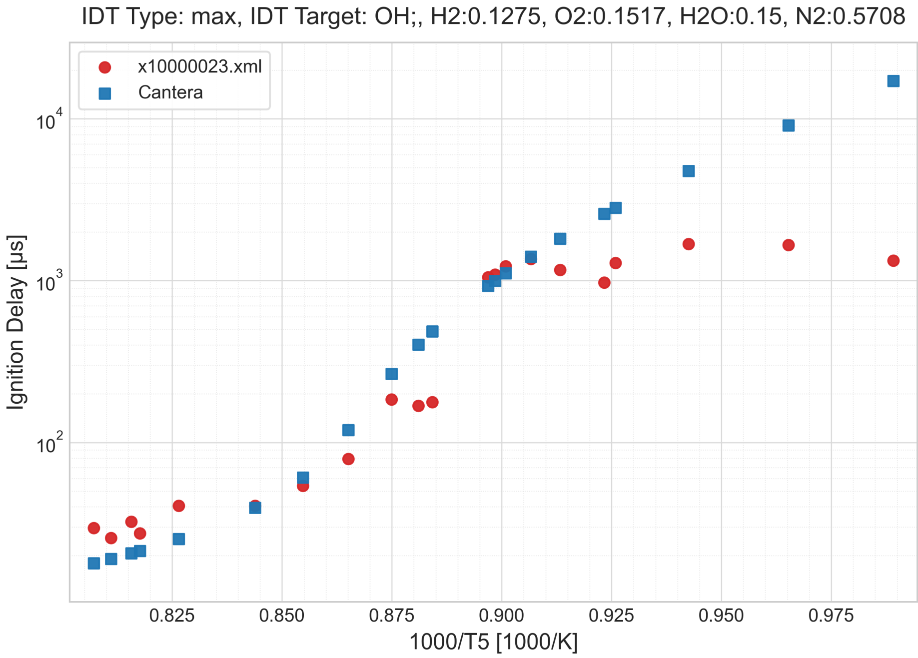


Figure 4.14: Comparison of Cantera IDT simulation with experimental measurement **good_cantera** IDT Type: OH max.

4.14 illustrates the discrepency between constant volume reactor Cantera simulation and the real shock tube experiment, where boundary layer buildup reduces the effective volume. This pressure rise should be compensated for in simulation to get more accurate IDT simulations. However in the RESPETCH syngas/hydrogen IDT datasets, pressure rise is rarely reported. It would be valuable for TUMKIN database to be more comprehensive.

4.5 Consistency Analysis

4.5.1 Sensitivity Analysis

4.5.2 Surrogate Model

4.5.3 B2BDC

4.5.3.1 Reaction Mechanisms

Reaction mechanisms used to validate the software developed in this thesis are Ethylene [18] and Combined Hydrogen Syngas [19]. Reported Chemkin input, transport and thermos files are converted to yaml by Cantera Chemkin to YAML conversion (c2kyaml).

5 Conclusion and Outlook

5.1 Outlook

A Appendix

B Program Flowcharts

References

C References

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Affirmation
Hereby, I affirm that I am the sole author of this thesis. To the best of my knowledge, I affirm that this thesis does not infringe upon anyone's copyright nor violate any proprietary rights. I affirm that any ideas, techniques, quotations, or any other material, are in accordance with standard referencing practices.

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Munich, March 2, 2026

YOUR NAME